MACHINE LEARNING PROJECT REPORT

Predicting Diabetes Disease Progression

Feature engineering, model complexity and regularization
A comparative study of OLS, Ridge and Lasso

Dataset	Development split	Selected model
442 patients / 10 features	353 training / 89 test	Lasso - original features

Abstract

This study predicts a continuous measure of diabetes disease progression one year after baseline, using the scikit-learn diabetes dataset. It compares ordinary least squares (OLS), Ridge and Lasso across the original variables, a manually engineered representation, and polynomial expansions of degrees 2 and 3. Exploratory analysis and tuning use the training partition; a fixed five-fold cross-validation rule selects the model before test evaluation.

Original-feature Lasso obtains the lowest selection-CV RMSE (55.3488), but its advantage over OLS (55.3946) is very small. The selected model achieves test RMSE 53.6857 and R^2 0.4560. Degree-2 Lasso has the lowest test error, but does not replace the pre-test recommendation. Unregularized cubic OLS severely overfits; rank diagnostics reveal redundant terms and numerical sensitivity. Regularization controls this expanded-model failure, while repeated nested validation does not establish a stability advantage for the tuned original-feature regularized models.

Central finding. Increasing feature complexity is useful only when its added flexibility is controlled. The strongest evidence for regularization is the prevention of polynomial-model failure and the production of sparse representations, rather than a decisive improvement over original-feature OLS.

1. Problem definition and dataset

The assignment concerns regression, not classification: the target is a quantitative disease-progression score approximately one year after the baseline measurements. The goal is to learn a mapping f: $R^{10} \rightarrow R$ from baseline characteristics to that score [1, 3]. The dataset contains 442 observations and ten predictors.

Feature	Description
age	Age
sex	Sex indicator
bmi	Body mass index
bp	Average blood pressure
s1	Total serum cholesterol
s2	Low-density lipoproteins
s3	High-density lipoproteins
s4	Total cholesterol / HDL ratio
s5	Log serum triglycerides
s6	Blood sugar level

Table 1. Baseline variables. The source representation is already centred and scaled; its numeric values are not measurements in original clinical units.

Data audit

Subset	Rows	Target mean	Target SD
Training	353	153.7365	78.0619
Test	89	145.7753	73.2008

Table 2. Split summaries recorded after model selection. Test-label summaries were not used to design the revised workflow.

There are no missing predictor or target values. Across all observations, the target ranges from 25 to 346, with mean 152.1335 and standard deviation 77.0930. These summaries describe the supplied dataset; they are not evidence of clinical causality.

Questions guiding the analysis

The study asks which characteristics are most associated with progression; how much variation a linear model explains; whether regularization helps with correlated predictors; whether nonlinear representations help; how conclusions depend on model choice; and which model balances accuracy with interpretability. The six answers are consolidated in Section 17.

2. Study design and evaluation protocol

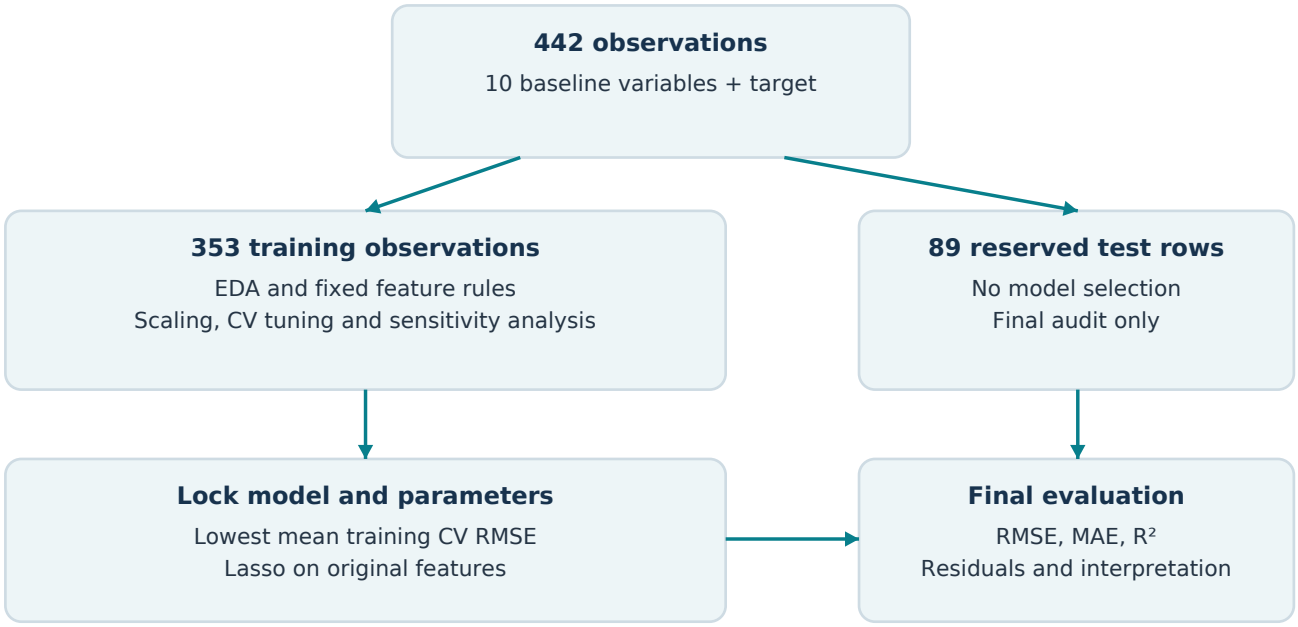


Figure 1. Evaluation workflow. The diagram is drawn from the current notebook's execution order; feature exploration and model selection precede test access.

Partitioning and model selection

A fixed 80:20 train/test split uses random_state = 42. Target-informed EDA is restricted to the 353 training rows. The feature rules are fixed before validation. Learned scaling is fitted inside each Pipeline, so validation folds do not determine the scaler. A shuffled five-fold KFold with seed 42 supplies the selection folds, and GridSearchCV minimizes mean RMSE. These choices follow the training/test separation required by the assignment [1, 6].

The model and parameters are locked before test predictions. The retained comparison loop reruns the same fixed searches; assertions verify agreement with the locked model and its predictions. A separate repeated nested-CV study uses training rows only and characterizes sensitivity of the three original-feature methods.

Performance measures

$$\text{RMSE} = \sqrt{[(1/n)\ \Sigma(y_i - \hat{y}_i)^2]}$$

$$\text{MAE} = (1/n)\ \Sigma|y_i - \hat{y}_i|$$

$$R^2 = 1 - \Sigma(y_i - \hat{y}_i)^2 / \Sigma(y_i - \bar{y})^2$$

RMSE is the primary selection metric and penalizes large errors. MAE gives an average absolute error in target units; R^2 compares prediction error with the variation around the evaluation-set mean. A negative R^2 indicates performance worse than that mean reference.

3. Exploratory analysis: distributions

Exploratory plots use the training partition only. The target occupies a broad range, so errors of several tens of progression-score units are plausible even when a model improves substantially on a mean predictor.

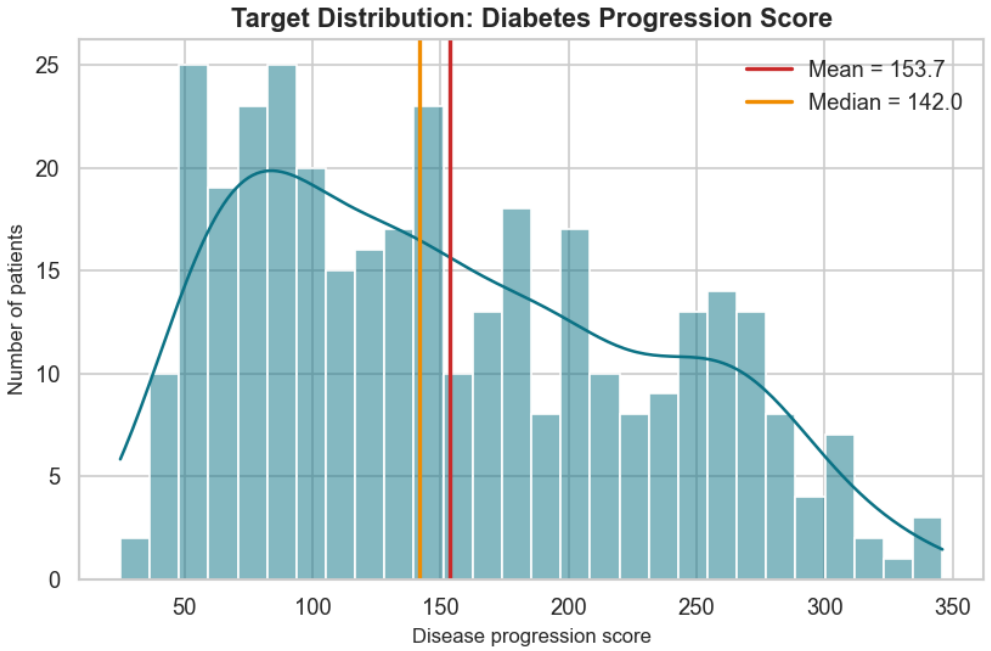


Figure 2. Distribution of the training target (n = 353), with its mean and median. Source: notebook cell 8.

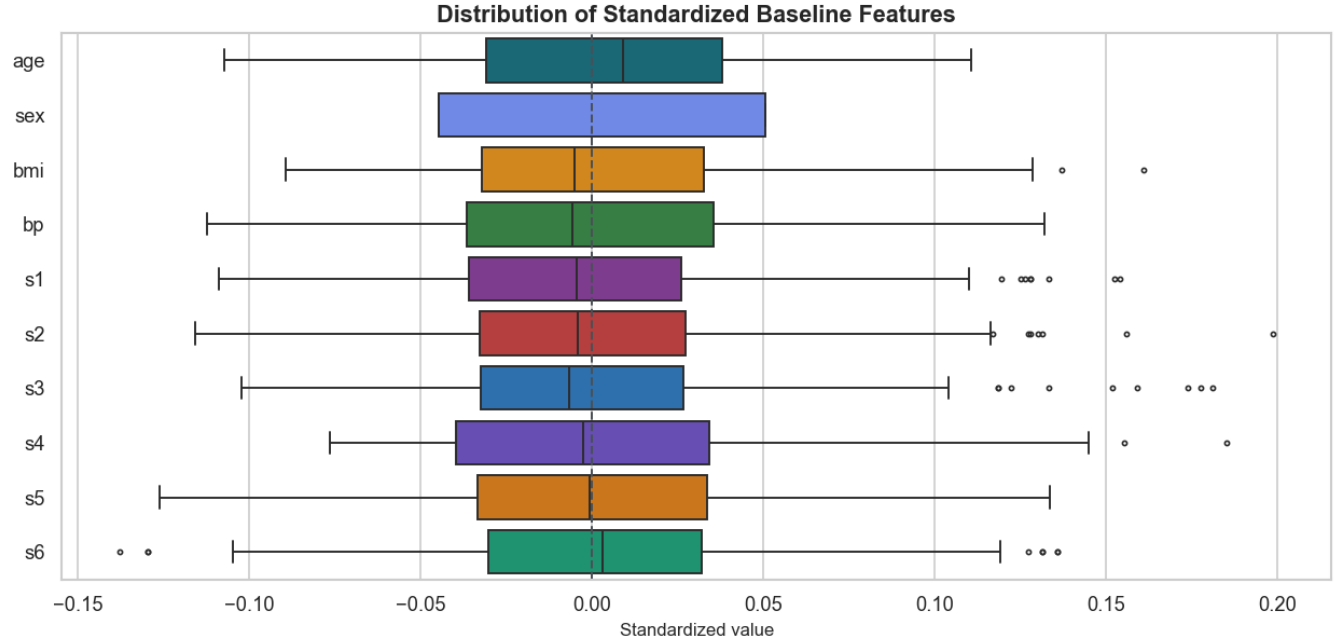


Figure 3. Distributions of the ten training predictors in the supplied scaled representation. Source: notebook cell 9.

The feature distributions differ in symmetry and tail length. The sex variable has two coded values, which matters when powers are later introduced. Extreme values are retained rather than removed automatically: an unusual baseline measurement is not, by itself, a data error.

4. Associations with disease progression

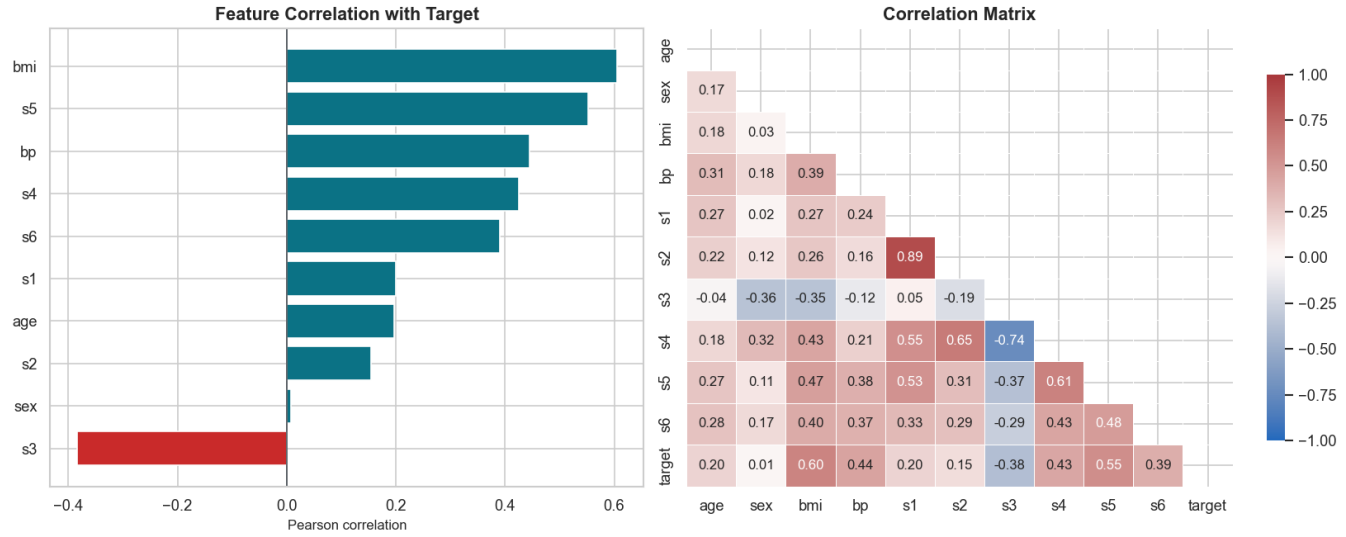


Figure 4. Training feature-target correlations and the correlation matrix. Source: notebook cell 11.

BMI has the strongest marginal association (r = 0.6048), followed by s5 (0.5522), blood pressure (0.4448), s4 (0.4251), and s6 (0.3904). HDL-related s3 is negatively associated (-0.3840). The near-zero marginal sex correlation (0.0071) does not imply that sex must have a zero conditional regression coefficient.

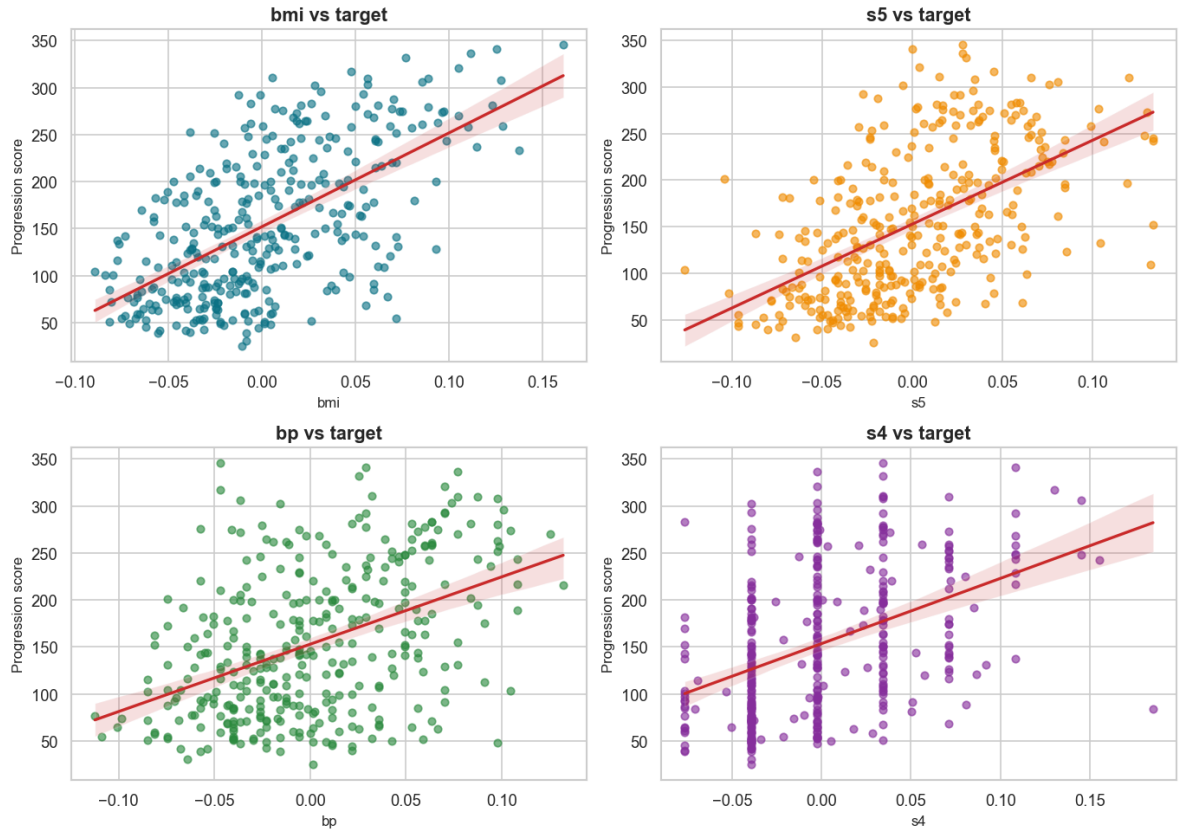


Figure 5. Scatter plots and fitted linear trends for the four strongest absolute training correlations. Source: notebook cell 12.

These plots indicate useful signal but considerable unexplained scatter. They show association, not causation. A high marginal correlation does not guarantee independent information once correlated predictors are included together.

5. Multicollinearity and feature design

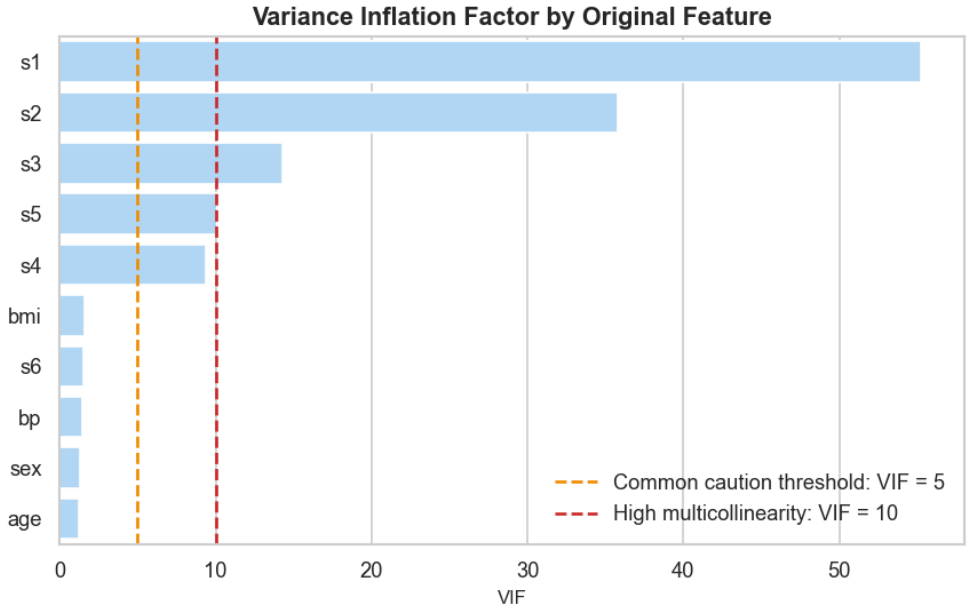


Figure 6. Training-set variance inflation factors (VIF). Source: notebook cell 14.

The strongest VIF values are s1 = 55.25, s2 = 35.76, s3 = 14.29, s5 = 10.07 and s4 = 9.33. These indicate that serum predictors contain substantial shared information. Large conditional coefficients and sign changes therefore require care; the marginal association plot alone cannot explain the fitted coefficient pattern.

Nine added manual features

Type	Terms	Purpose
Squares	bmi^2, bp^2, $s5^2$	Allow simple curvature
Interactions	bmi × bp, bmi × s5, bp × s5, s3 × s4	Allow joint effects
Lipid contrast	s1 - s3	Linear combination of existing variables
Metabolic composite	mean(bmi, bp, s5, s6)	Linear combination of scaled variables

Table 3. Manual representation: ten original predictors plus nine added terms, giving 19 columns.

The transformations are deterministic functions of predictors. They introduce no direct target values into the design matrix. However, choosing transformations after examining test-target relationships would still contaminate a strict holdout protocol; the revised workflow uses training data for this exploration.

The metabolic composite is a modelling feature, not a validated clinical risk score. The two linear composites add no independent linear information to OLS when their component variables are already included.

6. Representations and regression methods

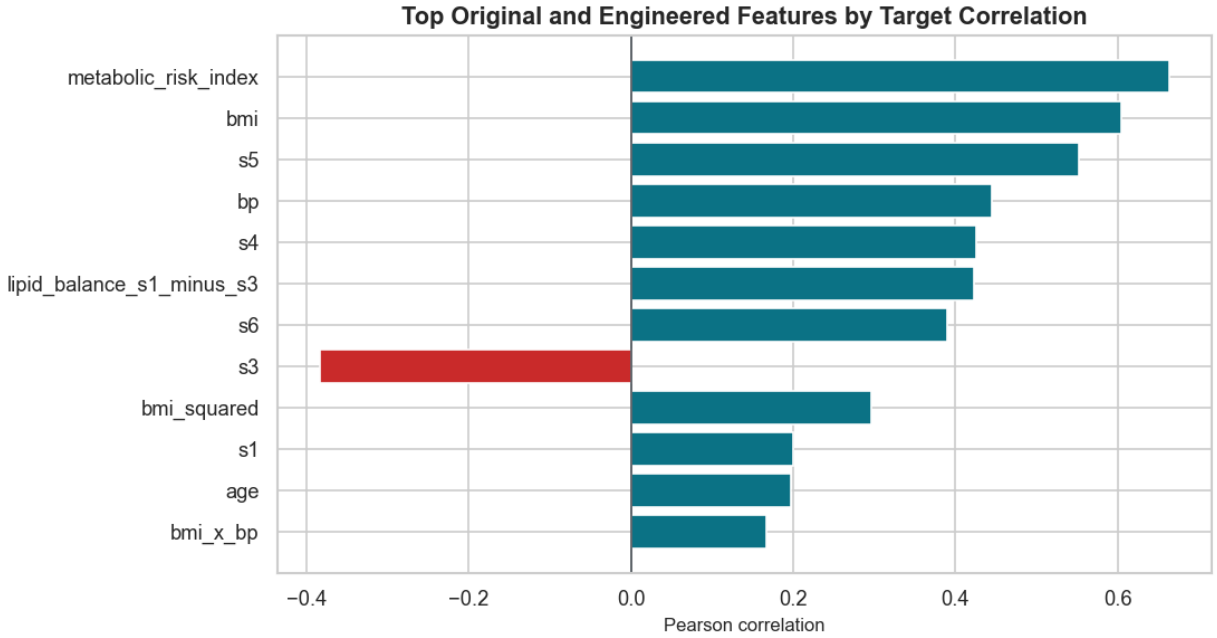


Figure 7. Strongest original/manual-feature correlations on training data. The metabolic composite has r = 0.6633. Source: notebook cell 18.

Representation	Columns	Examples
Original	10	age, bmi, bp, s1-s6
Manual	19	Original features + squares, interactions, composites
Polynomial degree 2	65	x^2 and pairwise interactions, plus linear terms
Polynomial degree 3	285	Cubic terms and interactions, plus lower degrees

Table 4. PolynomialFeatures uses include_bias=False; the regression estimator supplies the intercept [5].

What changes across OLS, Ridge and Lasso?

OLS minimizes squared residuals. **Ridge** adds an L2 penalty on coefficient size; **Lasso** adds an L1 penalty that can set coefficients exactly to zero. In the scikit-learn implementations, Ridge minimizes $||y - X\beta||^2 + \alpha||\beta||^2$, while Lasso minimizes $||y - X\beta||^2/(2n) + \alpha||\beta||_1$. Their α values therefore should not be compared as though they shared identical normalization [4].

The intercept is not included in the polynomial expansion or the reported feature counts. Each representation is scaled inside the model pipeline. Coefficients are interpreted in the fitted standardized feature units, with comparisons primarily made between methods using the same representation.

7. Cross-validated model comparison

All candidates use the same training partition and five selection folds. The primary ranking uses mean validation RMSE. Fold SD measures variation across folds; it is not a confidence interval or a test of statistical significance.

Model	CV RMSE	Fold SD	Train RMSE	Gap
Lasso - original	55.35	2.35	53.41	1.94
Elastic Net - original	55.37	2.37	53.38	1.99
Ridge - original	55.37	2.37	53.37	2.00
OLS - original	55.39	2.36	53.35	2.05
OLS - manual	55.42	2.56	52.11	3.30
Ridge - manual	55.42	2.55	52.11	3.30
Lasso - manual	55.42	2.55	52.11	3.31
Lasso - degree 2	55.97	1.91	52.92	3.05
Gradient boosting	56.63	2.30	46.50	10.13
Ridge - degree 2	56.89	1.67	50.99	5.90
Lasso - degree 3	57.34	2.99	54.93	2.41
Random forest	57.75	1.85	44.10	13.64
Ridge - degree 3	59.33	3.39	50.18	9.15
OLS - degree 2	62.50	4.41	47.63	14.87
Mean baseline	78.13	5.64	77.90	0.24
OLS - degree 3	2216.92	1381.78	10.83	2206.09

Table 5. Saved selection-CV results for all 16 candidates. Gap = mean validation RMSE minus mean training RMSE. Source: notebook cell 26.

Locked recommendation: Lasso on original features, α = 0.1467799268. Its CV RMSE is 55.3488, compared with 55.3946 for OLS and 55.3742 for Ridge. The Lasso-OLS difference is only 0.0458, so the numerical ranking should not be interpreted as decisive superiority.

Manual feature models perform similarly to original-feature regressions. Degree-2 Lasso is the best polynomial model by selection CV (55.9712). Cubic OLS has a very low training error but an enormous validation error, signalling failure of generalization. Ensemble models provide supplementary nonlinear comparisons rather than replacing the required OLS/Ridge/Lasso study.

8. Final test evaluation

After the selection rule is fixed, test predictions are computed for the predefined candidates. The test table is a descriptive audit. Its lowest-error row does not replace the model selected using training CV.

Model	Test RMSE	Test MAE	Test R^2
Lasso - degree 2	50.69	41.27	0.5151
Ridge - degree 2	51.34	41.19	0.5025
Gradient boosting	52.70	41.61	0.4758
Lasso - degree 3	52.94	43.89	0.4710
Lasso - manual	53.01	41.96	0.4696
Ridge - manual	53.01	41.96	0.4696
OLS - manual	53.01	41.96	0.4695
Random forest	53.05	42.71	0.4688
Lasso - original	53.69	42.82	0.4560
Elastic Net - original	53.74	42.81	0.4549
Ridge - original	53.78	42.81	0.4541
OLS - original	53.85	42.79	0.4526
Ridge - degree 3	55.40	45.81	0.4206
OLS - degree 2	55.64	43.58	0.4156
Mean baseline	73.22	64.01	-0.0120
OLS - degree 3	287.13	164.85	-14.5613

Table 6. Saved test metrics for 89 observations. The highlighted row is the CV-selected model. Source: notebook cell 26.

Accuracy and model choice

The selected Lasso achieves RMSE 53.6857, MAE 42.8152 and R^2 0.4560. It improves substantially on the mean baseline, whose test RMSE is 73.2225 and R^2 is -0.0120. The selected model captures useful predictive variation, but much remains unexplained.

Degree-2 Lasso has the lowest test RMSE (50.6853) and highest listed test R^2 (0.5151). This suggests a potentially useful nonlinear representation in this split. Nevertheless, its selection-CV score is worse than original-feature Lasso, and the test set is too small to establish that the test ranking will persist in new data.

9. Polynomial degree and overfitting

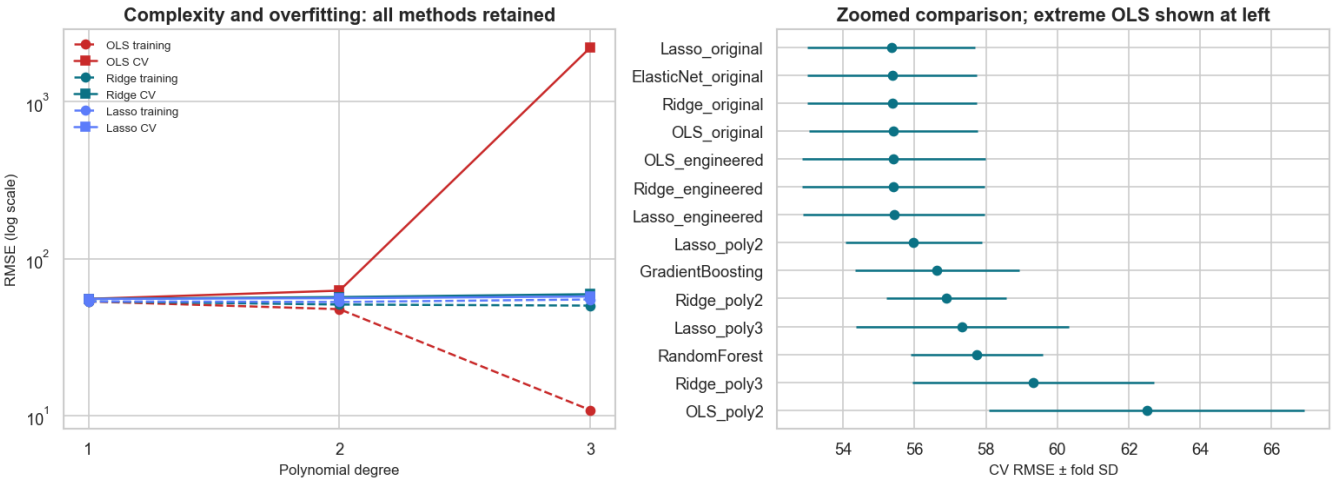


Figure 8. Training versus validation error by degree (log scale), with a readable zoom of candidate CV scores. Error bars are fold SD. Source: notebook cell 43.

Degree	Method	Terms	Active	Train RMSE	CV RMSE	Test RMSE
1	OLS	10	10	53.35	55.39	53.85
1	Ridge	10	10	53.37	55.37	53.78
1	Lasso	10	9	53.41	55.35	53.69
2	OLS	65	65	47.63	62.50	55.64
2	Ridge	65	65	50.99	56.89	51.34
2	Lasso	65	15	52.92	55.97	50.69
3	OLS	285	285	10.83	2216.92	287.13
3	Ridge	285	285	50.18	59.33	55.40
3	Lasso	285	9	54.93	57.34	52.94

Table 7. Degree comparison for the three required methods. “Active” means |coefficient| > 1e-8. Source: notebook cell 43.

For OLS, increasing degree lowers training RMSE from 53.35 to 47.63 and then 10.83, while validation RMSE increases to 62.50 and then 2216.92. This is the clearest overfitting result. Ridge and Lasso greatly reduce the failure at degree 3, but higher degree still does not improve their selection-CV scores over the original representation.

10. Redundancy and numerical stability

Representation	Columns	Rank	Redundant	Condition no.
Original	10	10	0	2.07e+01
Manual engineered	19	17	2	8.86e+15
Polynomial degree 2	65	64	1	7.59e+15
Polynomial degree 3	285	274	11	2.87e+16

Table 8. Training design diagnostics after pipeline transformation and centring. The default numerical rank is based on a floating-point tolerance.

Degree 3 creates 285 columns, whereas an individual five-fold CV training fit has only 282 or 283 observations. The full training matrix has numerical rank 274. Some dependencies arise because sex has only two encoded values: if they are a and b, then $sex^2 = (a+b)sex - ab$. Higher powers and related interactions also create redundant columns.

The manual matrix has 19 columns but rank 17 because its lipid contrast and metabolic composite are linear combinations of included predictors. Rank deficiency makes individual OLS coefficient allocations non-unique or numerically fragile; a low residual fitting error does not resolve that problem.

Penalty settings and coefficient magnitudes

Set	Ridge α	Lasso α	OLS $\|\|\beta\|\|_2$	Ridge $\|\|\beta\|\|_2$	Lasso $\|\|\beta\|\|_2$
Original	1.0000	0.1468	71.6	60.9	53.5
Manual	0.0010	0.0010	295	294	294
Degree 2	81.6140	3.1623	1.31e+03	36.7	36.3
Degree 3	417.5319	6.8129	1.37e+06	19	32

Table 9. Selected penalties and coefficient norms from the saved notebook. Norm comparisons are meaningful primarily within a representation.

The cubic OLS norm is approximately 1.37 million, compared with 19.05 for Ridge and 32.01 for Lasso. This large difference illustrates the role of regularization in controlling coefficient magnitude. It does not by itself prove which model predicts best; the validation table supplies that evidence.

Reproducibility limitation. The notebook records cubic OLS CV/test RMSE as 2216.9178/287.1342 and notes a prior environment's values of 1402.1769/278.9423. The qualitative failure is consistent, but the precise large error is numerically sensitive. This report uses the saved run throughout.

11. Coefficient selection across methods

Model	Terms	Active	Zero	Largest |coefficient| term
OLS - original	10	10	0	s1
Ridge - original	10	10	0	s1
Lasso - original	10	9	1	s5
OLS - manual	19	19	0	s2
Ridge - manual	19	19	0	s2
Lasso - manual	19	17	2	s2
OLS - degree 2	65	65	0	s1
Ridge - degree 2	65	65	0	bmi
Lasso - degree 2	65	15	50	bmi
OLS - degree 3	285	285	0	s1 s2
Ridge - degree 3	285	285	0	bmi
Lasso - degree 3	285	9	276	sex^2 bmi

Table 10. All 12 required regression/representation combinations. A zero threshold of 1e-8 is used. Source: notebook cell 41.

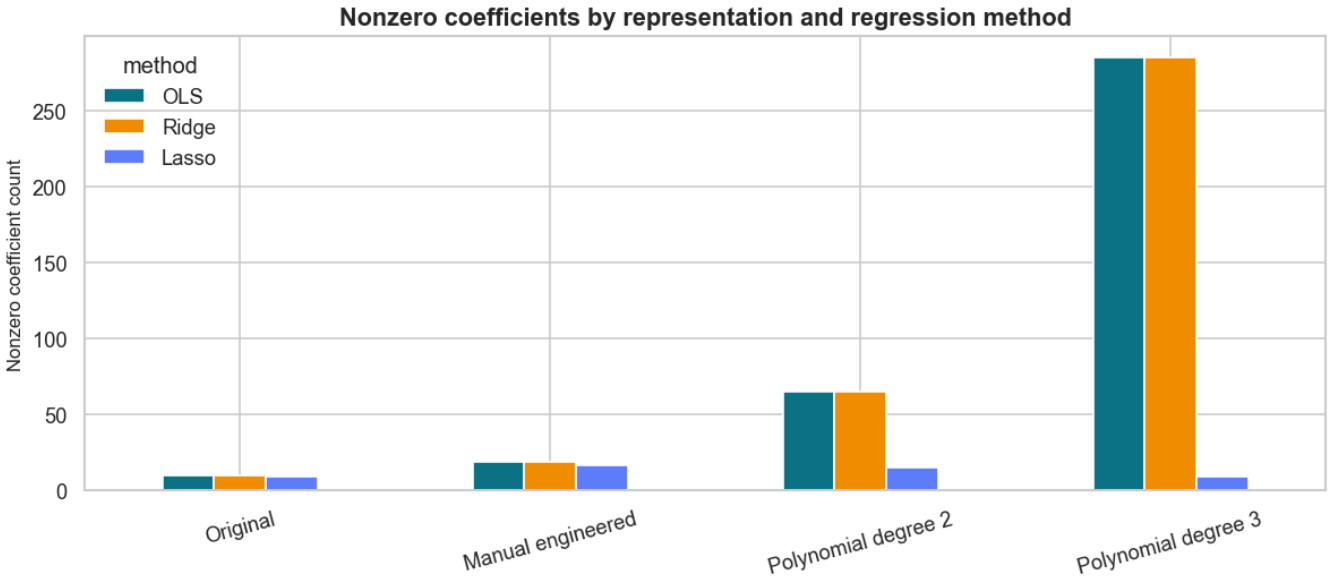


Figure 9. Nonzero coefficient counts for OLS, Ridge and Lasso across the four representations. Source: notebook cell 41.

OLS and Ridge retain all displayed coefficients above the numerical threshold, whereas Lasso retains 9/10 original, 17/19 manual, 15/65 quadratic and 9/285 cubic terms. The sparsity is representation-dependent: polynomial terms containing sex can be proxies for other included columns because of binary-feature identities.

12. Interpreting the fitted coefficients

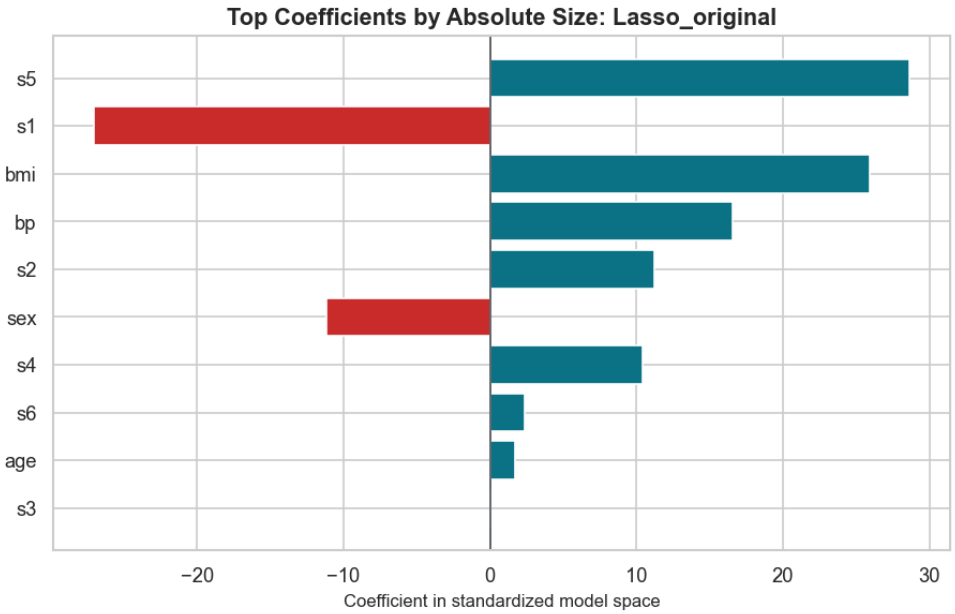


Figure 10. Coefficients of the CV-selected original-feature Lasso in standardized model units. Source: notebook cell 39.

Feature	OLS	Ridge	Lasso
s1	-44.449	-34.672	-27.014
s5	35.161	31.378	28.653
bmi	25.607	25.733	25.912
s2	24.641	17.053	11.208
bp	16.829	16.734	16.591
s4	13.139	11.764	10.400
sex	-11.512	-11.448	-11.193
s3	7.677	3.370	-0.000
s6	2.351	2.458	2.393
age	1.754	1.807	1.692

Table 11. Original-feature coefficients, refitted on the full training partition. All methods use the same representation and scaling.

The selected Lasso gives positive weights to BMI, s5 and blood pressure and sets s3 to zero. The s1 coefficient is negative despite its positive marginal association with the target. Such differences reflect conditional relationships among correlated predictors; they do not imply causal effects or invalidate the marginal EDA.

A zero Lasso coefficient means the term was unnecessary under this representation, penalty and training sample. It does not prove that the underlying baseline characteristic is biologically irrelevant. Likewise, coefficient magnitude is not a universal importance score across different feature spaces.

13. What do expanded models retain?

Largest terms across all three methods

Representation	Method	Three largest absolute coefficients (signed values)
Manual	OLS	s2 (198); s1 (-156); lipid_balance_s1_minus_s3 (-108)
Manual	Ridge	s2 (198); s1 (-155); lipid_balance_s1_minus_s3 (-107)
Manual	Lasso	s2 (197); s1 (-163); s5 (102)
Degree 2	OLS	s1 (-811); s2 (705); s1 s2 (-446)
Degree 2	Ridge	bmi (18.5); s5 (17); bp (14.7)
Degree 2	Lasso	bmi (24.1); s5 (19.6); bp (13.5)
Degree 3	OLS	s1 s2 (-9.49e+05); s1^2 (6.16e+05); s2^2 (3.78e+05)
Degree 3	Ridge	bmi (5.28); sex^2 bmi (5.23); s5 (4.85)
Degree 3	Lasso	sex^2 bmi (23.6); s5 (17.8); sex^2 bp (10.2)

Table 12. Leading terms calculated from the complete coefficient tables stored in notebook cell 41. They are not causal rankings.

All retained polynomial Lasso terms

Degree-2 term	Coefficient	Degree-3 term	Coefficient
bmi	24.1280	sex^2 bmi	23.6494
s5	19.6110	s5	17.8176
bp	13.4714	sex^2 bp	10.1918
s3	-10.0567	s3	-5.9740
sex^2	-5.4557	bmi^3	2.6347
age sex	3.9645	age^2 sex	-0.7498
bmi^2	3.2153	bp^3	0.6550
bmi bp	2.7200	s3 s6^2	-0.5309
s6^2	1.8841	bmi^2	0.0010
age bp	1.0667		
age s5	1.0556		
s6	0.8261		
age s2	-0.6616		
bmi s6	0.2125		
sex	-0.0027		

Table 13. All 15 quadratic and 9 cubic active Lasso terms. Near-zero active terms remain counted under the stated 1e-8 threshold.

A retained expression such as sex² × bmi must be interpreted in the context of the coded binary feature and correlated polynomial basis. Selection of a particular basis term is not evidence of a unique physiological interaction.

14. Repeated nested validation and stability

The three original-feature methods are evaluated with five outer folds repeated three times on training data. Ridge and Lasso are retuned with three-fold inner CV in each outer training subset. All methods share the outer splits, producing 15 scores per method and 45 model scores in total. The final test rows never enter this study.

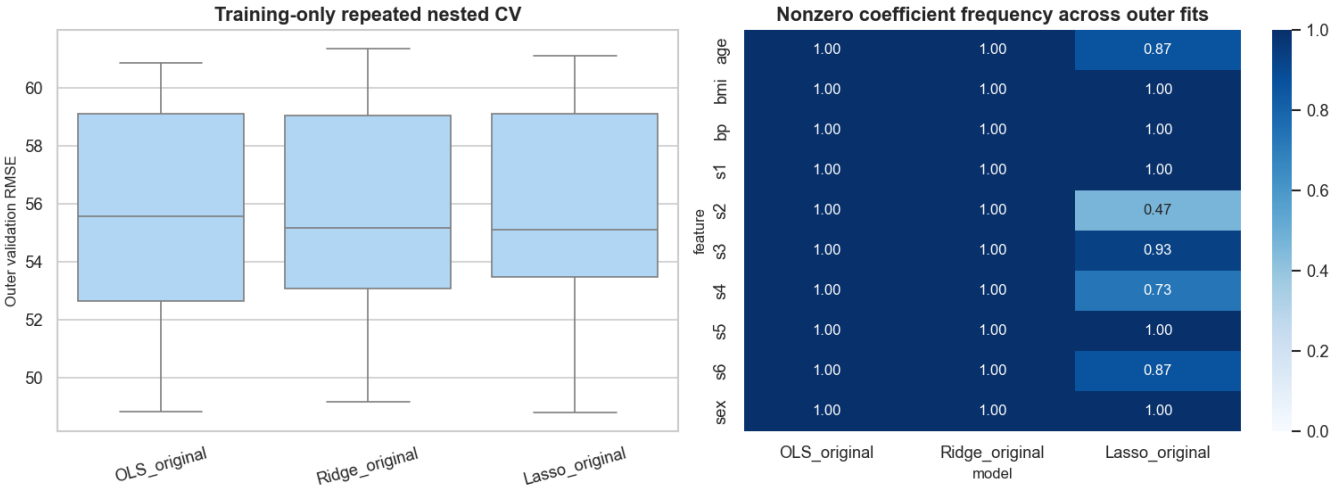


Figure 11. Outer validation error distributions and nonzero-coefficient frequencies. Coefficient variability is measured after conversion to common training-standard-deviation units. Source: notebook cell 25.

Method	Mean outer RMSE	Outer RMSE SD	Mean coefficient SD
OLS_original	55.4741	3.8852	4.1765
Ridge_original	55.5969	3.8428	6.0305
Lasso_original	55.7510	3.8580	6.2182

Table 14. Repeated nested validation and coefficient sensitivity on the training partition.

Paired comparison	Mean ΔRMSE	Folds better than OLS
Ridge_original minus OLS	0.1228	6/15
Lasso_original minus OLS	0.2769	6/15

Table 15. Positive ΔRMSE indicates higher error than OLS. The folds overlap across repeats and are not independent experiments.

In this experiment, retuned Ridge and Lasso do not improve mean outer error or reduce average coefficient variability relative to OLS. The earlier general expectation that regularization stabilizes coefficients is therefore not an empirical conclusion for these tuned original-feature procedures. Penalty selection itself can introduce variability across resamples.

The score differences are small relative to the observed variation. Fold SDs and win counts are descriptive; no naive significance test or independent-sample confidence interval is asserted. This nested study covers the original-feature methods, not every expanded candidate.

15. Diagnostics of the selected model

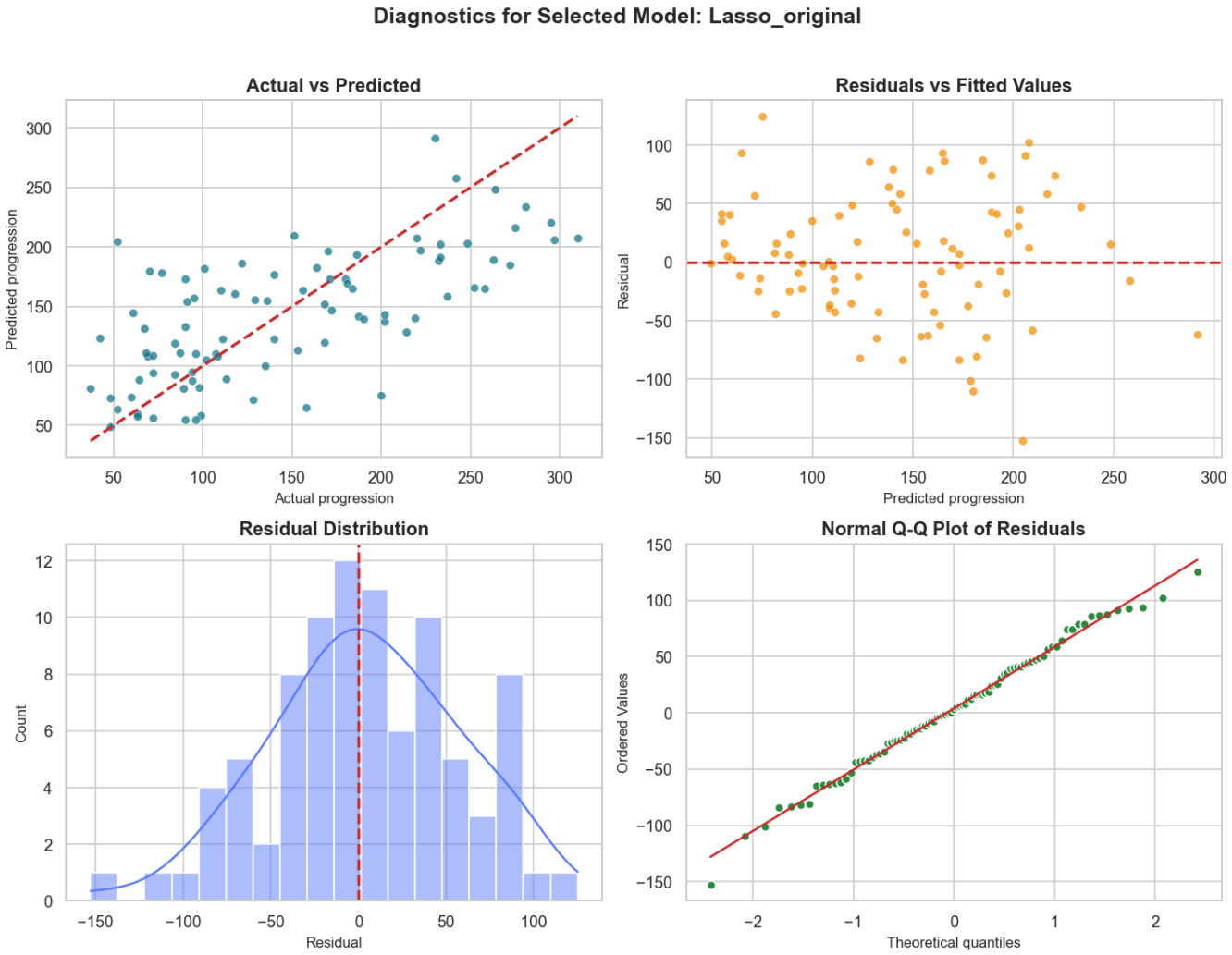


Figure 12. Selected Lasso: actual versus predicted, residuals versus fitted values, residual distribution and normal Q-Q plot. Residual = actual - predicted. Source: notebook cell 35.

Actual-target band	Cases	Mean actual	Mean predicted	Mean residual
lowest quarter	23	65.26	104.00	-38.74
middle-low	22	103.73	118.83	-15.11
middle-high	23	169.30	148.40	20.91
highest quarter	21	252.24	200.19	52.05

Table 16. Post-selection residual summary by test-target quartile. Unequal counts arise from ties in target values.

The lowest target quarter is overpredicted on average by 38.74 units, whereas the highest quarter is underpredicted by 52.05 units. Predictions are pulled toward the centre of the observed range. This pattern helps explain why useful aggregate R^2 coexists with substantial errors for individual cases.

The residual and Q-Q plots are diagnostic descriptions, not proof that all model assumptions hold. Normal residuals are not required simply to calculate RMSE. These findings are not used to retune against the test set.

16. Feature importance and error analysis

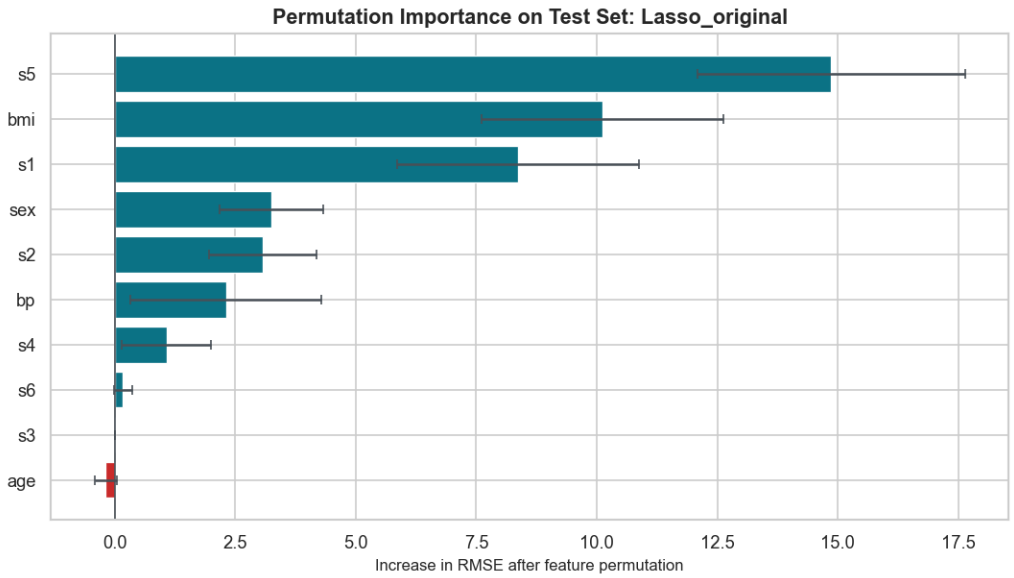


Figure 13. Permutation importance of the selected model on the test set: increase in RMSE after shuffling each feature. Error bars describe variation over 30 permutations. Source: notebook cell 44.

The largest permutation effects are associated with s5, BMI and s1. Because predictors are correlated, permutation scores and coefficients need not agree with marginal correlations. These post-selection scores are explanatory; no features are removed in response to them.

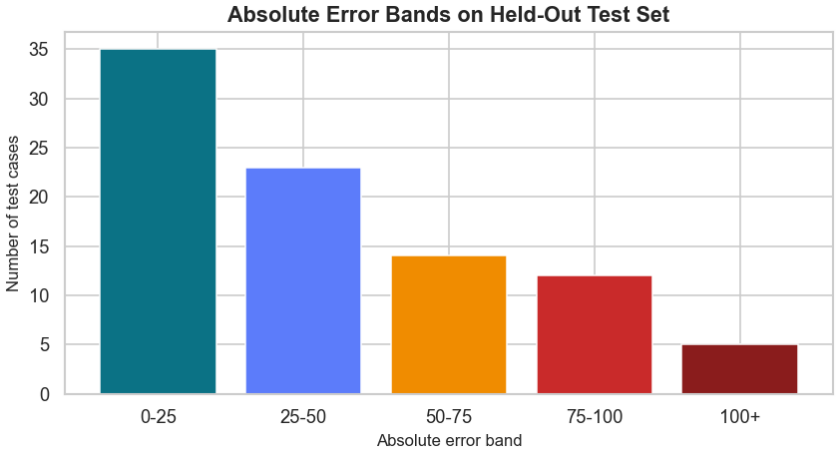


Figure 14. Absolute-error bands for the 89 test observations. Source: notebook cell 47.

Dataset row	Actual	Predicted	Absolute error
56	52.0	204.72	152.72
77	200.0	75.10	124.90
211	70.0	179.99	109.99
9	310.0	207.70	102.30
209	77.0	178.41	101.41

Table 17. Five largest test errors. Row numbers are dataset indices, not patient identifiers.

Thirty-five cases have absolute error at most 25, while five exceed 100. The largest error is 152.72. These errors show why this educational model should not be treated as an individual clinical decision system.

17. Answers to the research questions

1. Which baseline characteristics are most associated with progression?

Training correlations are strongest for BMI (0.6048), s5 (0.5522), blood pressure (0.4448), s4 (0.4251), s6 (0.3904) and s3 (-0.3840). Correlated serum predictors can have different marginal and conditional roles. These findings establish association within this dataset, not causation.

2. How well does a linear model explain target variation?

Original OLS achieves test RMSE 53.8534, MAE 42.7941 and R^2 0.4526, compared with mean-baseline RMSE 73.2225. This is useful predictive signal, but more than half the test variation remains unexplained under the R^2 comparison. “Explains” here refers to predictive fit, not a causal mechanism.

3. Does regularization improve generalization with correlated predictors?

Original-feature selection CV only slightly favours Lasso over OLS. Repeated nested validation does not show better mean performance or smaller coefficient variability for the tuned original-feature regularized procedures. The strongest observed benefit is control of the severe polynomial OLS failure and, for Lasso, sparse coefficient selection.

4. Do nonlinear models capture additional relationships?

The degree-2 Lasso test score is better than the original-feature scores, but its selection-CV score is worse. Manual features and ensembles also fail to demonstrate a decisive training-CV advantage. Nonlinear signal is possible, but complexity increases variance and the test ranking is not sufficient evidence of a reliable improvement.

5. How sensitive are conclusions to model choice?

The exact prediction ranking, active terms and coefficient allocations vary with model and representation. Correlated terms can exchange weight, and the CV winner differs from the test winner. The repeated coefficient frequencies and degree comparisons make this sensitivity visible; a single coefficient table would not establish stable predictor rankings.

6. Which model best balances accuracy and interpretability?

The fixed selection rule chooses original-feature Lasso. It uses nine active coefficients and obtains test RMSE 53.6857 and R^2 0.4560. It is a reasonable compact recommendation, while original OLS and Ridge remain competitive. Its small CV advantage is insufficient to claim a uniquely superior model; the choice is conditional on the declared procedure.

18. Conclusions and limitations

Conclusions

The study meets the central objective by comparing representation and regularization jointly. Increasing polynomial degree expands the hypothesis space and can sharply reduce training error without improving validation performance. With 285 degree-3 terms and a rank-deficient design, cubic OLS is especially vulnerable. Its failure is informative evidence about model complexity, not a reason to hide an unfavourable result.

Ridge constrains coefficient magnitude, while Lasso also removes terms. Their expanded representations behave far more reasonably than cubic OLS. On the original variables, however, all three regression methods perform similarly. The added nested study qualifies the expectation of improved stability: it does not support that claim for the tuned original-feature models in this experiment.

Final recommendation. Retain the pre-test choice of original-feature Lasso, report OLS and Ridge as competitive alternatives, and treat degree-2 Lasso's test advantage as a descriptive finding requiring further confirmation. Prefer a transparent explanation of uncertainty over a claim that a marginal ranking identifies a universally best model.

Limits of the evidence

Sample and holdout history. There are only 442 observations and 89 test cases. The dataset and test split were inspected in an earlier notebook version. The revised training-only workflow is explicit, but it cannot retroactively create a pristine external holdout. A new independent dataset or prospectively untouched evaluation would be needed for stronger confirmation.

Model-selection uncertainty. The selection-CV minimum is used to choose among many candidates and can be optimistic as a performance estimate. Nested validation is provided for the three original-feature methods only. The repeated outer scores overlap; their variability is not an independent-sample confidence interval.

Representation and interpretation. High serum-feature correlations and redundant polynomial terms complicate coefficient interpretation. A sparse basis is not a unique causal explanation. The supplied scaled variables and constructed composites should not be interpreted directly as clinical quantities or risk scores.

Numerical and practical reliability. The cubic OLS result is sensitive to numerical environment. The notebook documents versions and exposes solver warnings, but the analysis remains an educational regression exercise. It has not been validated as a clinical deployment system.

19. Reproducibility and references

Recorded notebook environment

Component	Recorded version
python	3.13.12
scikit-learn	1.8.0
numpy	2.4.2
scipy	1.17.0
pandas	3.0.0
matplotlib	3.10.8
seaborn	0.13.2

Table 18. Versions from the current saved notebook run, not the separate environment used to typeset this report.

The notebook contains 58 cells, including 35 code cells with saved execution counts 36-70 and no saved error outputs. Cell references count all Markdown and code cells from 1; they are not execution counts. random_state = 42 defines the main split and selection folds. The repeated outer study uses seed 142; inner folds use 42 + outer-fold number. These details support reproduction of the analysis, subject to numerical sensitivity of the failed cubic OLS fit.

References

[1] Assignment specification. Project: Predicting Diabetes Disease Progression. ePDG-ML-Project1.pdf, supplied course assignment, sections 1-6.

[2] Companion notebook. diabetes_regression_project_detailed.ipynb. Current saved analysis, figures and model outputs supplied with this report.

[3] scikit-learn: Diabetes dataset. scikit-learn.org - toy datasets / diabetes

[4] scikit-learn: Linear models. OLS, Ridge and Lasso objectives. scikit-learn.org - linear models

[5] scikit-learn: Polynomial features. scikit-learn.org - preprocessing / polynomial features

[6] scikit-learn: Data leakage. scikit-learn.org - common pitfalls / data leakage